

Assignment #02

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
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Due Date: Friday October 10th, my Mailbox (3:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name(s) in the report. Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
 - Bring a hard copy of your report as a PDF file by the Due Date and Time in class

 - Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report
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Problem #1

In order to calculate the optimal water flow rate the optimizer in Aspen was utilized. The given operating conditions of 25°C and a vapor fraction of 0 for cooling water were specified. The temperature of 57.0 C and flow rate of 2088 kmol/hr of propane was specified in the Thot in. Inside the heat exchanger the outlet hot stream vapor fraction of 0 was specified and the given heat transfer coefficient of 425.6 W/ (m² * K).

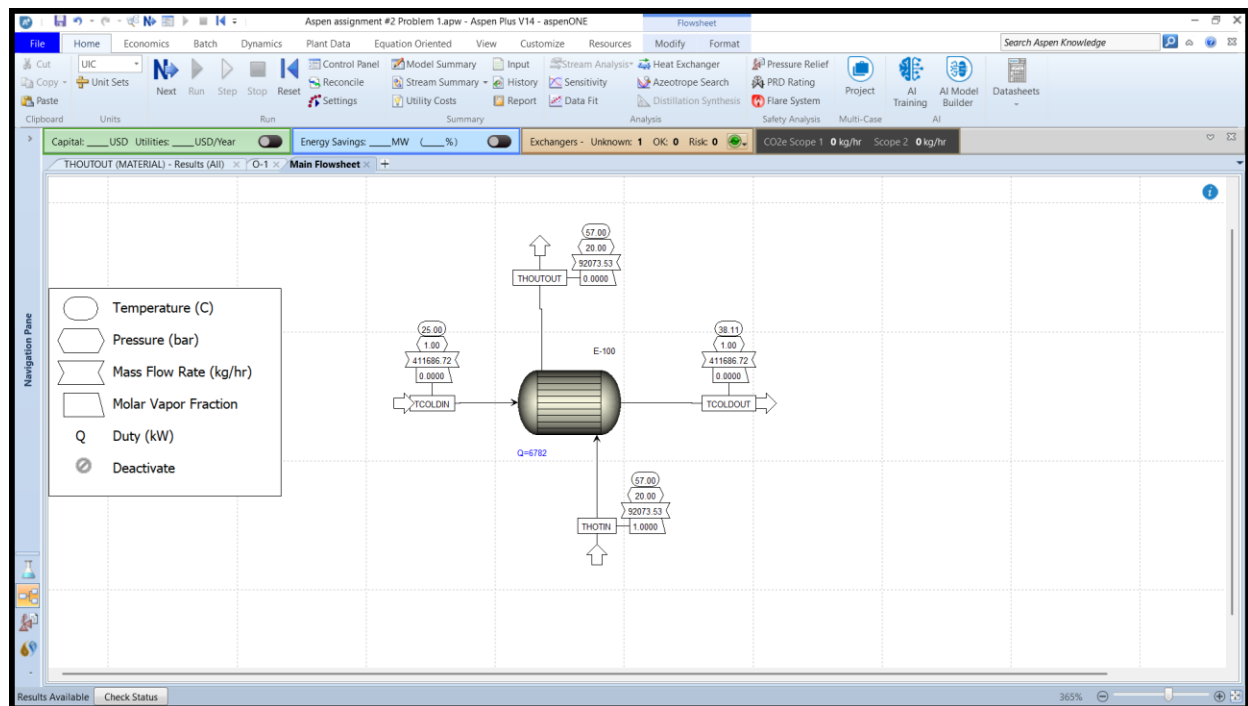


Figure 1 PFD for problem #1

Following the solution the heat and weight balance is as shown in figure 2

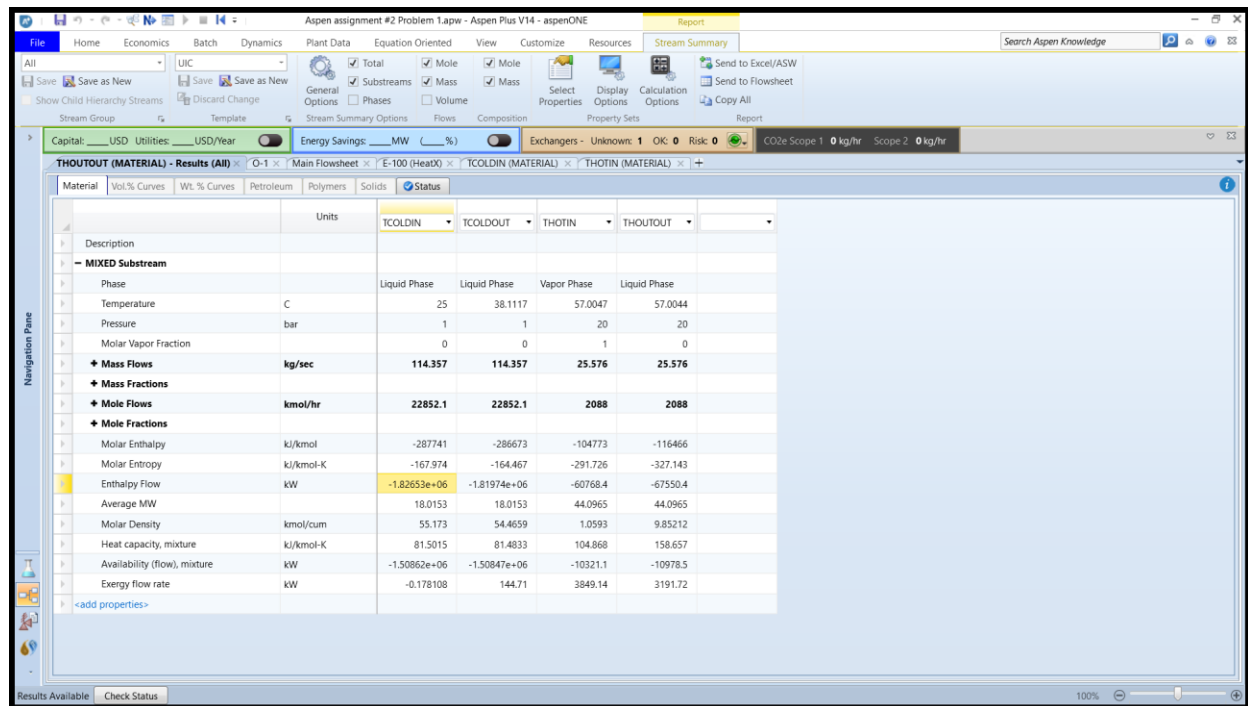


Figure 2 Heat and Weight Balance #1

1-3 Solution

For the solution to the problem the ASPEN optimizer was utilized. With the operating conditions given from the problem statement a calculator was utilized. This calculator was used to calculate the total cost utilizing the formulas from Gavin's book which were given in the problem statement. All the given stream variables were imported as import variables and the values for purchase cost and operating cost were exported as export variables. Since ASPEN calculates the area for the heat exchanger it can also be imported as an import variable and utilized in the calculation of the cost. The costs were calculated and labeled as a variable CTOT.

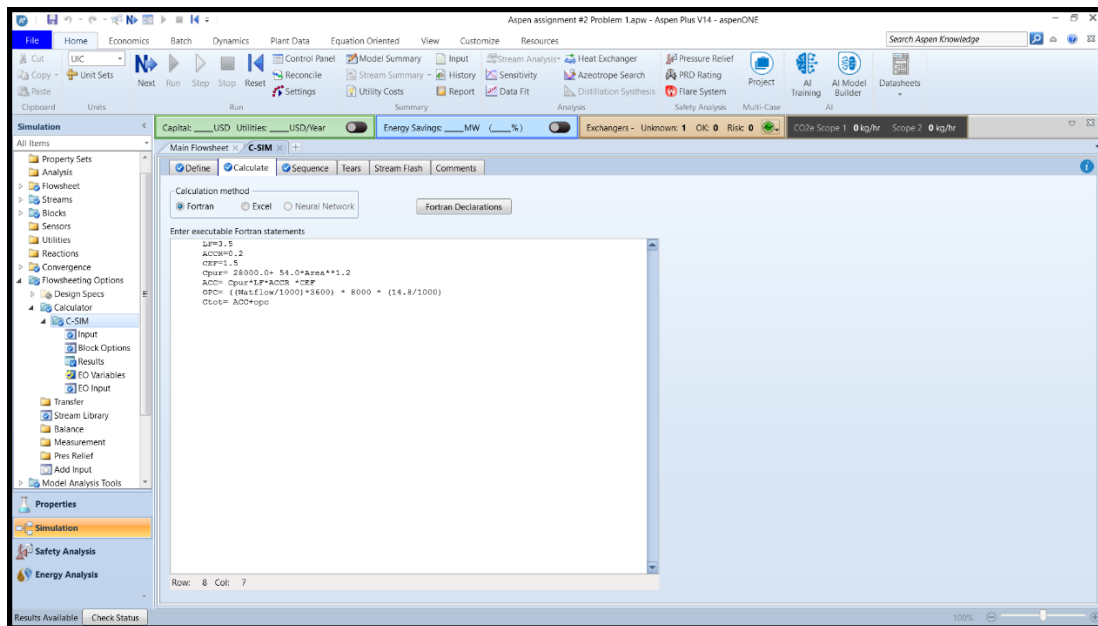


Figure 3 Calculator CODE

This cost was exported as an objective to the optimizer. The optimizer was used to minimize the total cost by changing the incoming mass flow rate subject to the constraint that the temperature rise could not be more than 20°C. This was done to ensure the solution converged and is a reasonable assumption that we have made many times in class. The result is that the optimal flow rate of water is 114.357 kg/sec with a total yearly cost of \$187,323.

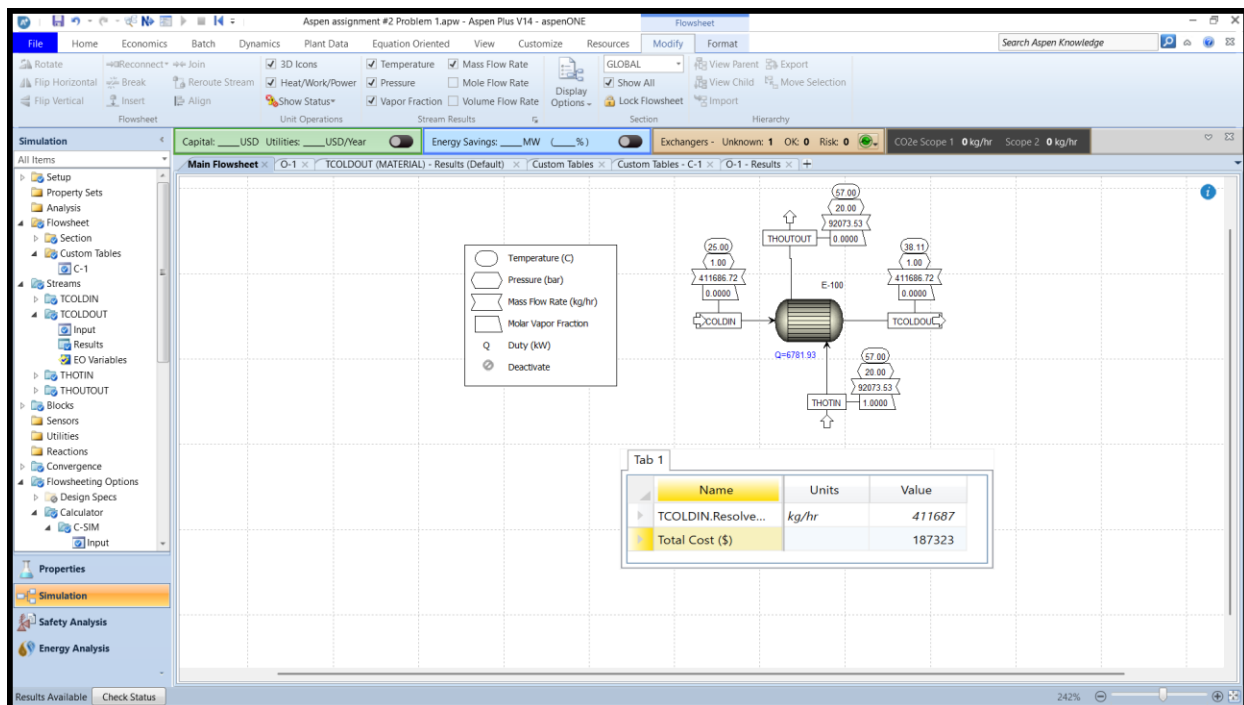


Figure 4 Final Results

Problem #2

2.1 PFD

To compare the costs between a three and four stage refrigeration loop both were set up in ASPEN. The three-stage refrigeration loop is as shown:

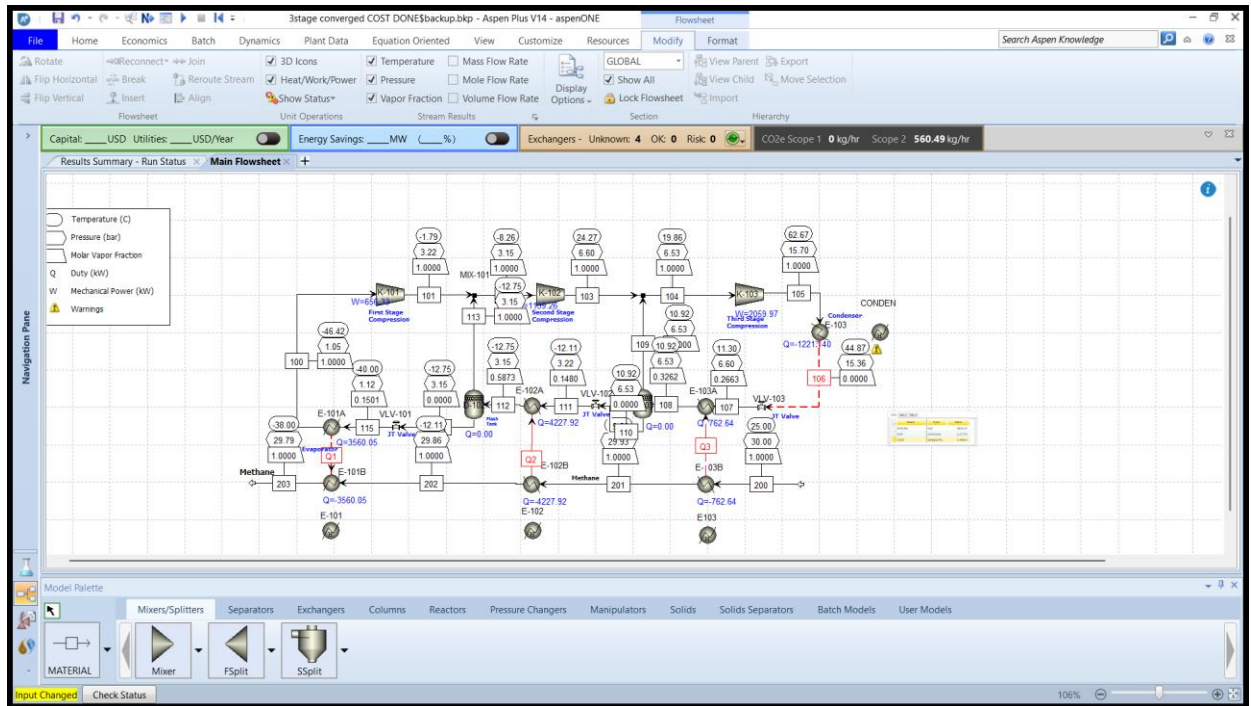


Figure 5 3 Stage Refrigeration Loop

To accomplish the converged flowsheets various methods within ASPEN were utilized. A geometric mean of the pressure initial values and final values was used in an initial guess for the pressures out of the valve and it was then exported to the outlet of each compressor the stream would mix with. A design spec was utilized that changed the mass flow rate of the tear stream to achieve the target outlet temperature of -38°C . Once this was done the water necessary for the condenser was calculated by using another design spec that changed the flow rate of cooling water to achieve the given temperature change of 10°C in the condenser. Following this

the flowsheet was fully converged. The heat and weight balance for the three-stage compression is as follows:

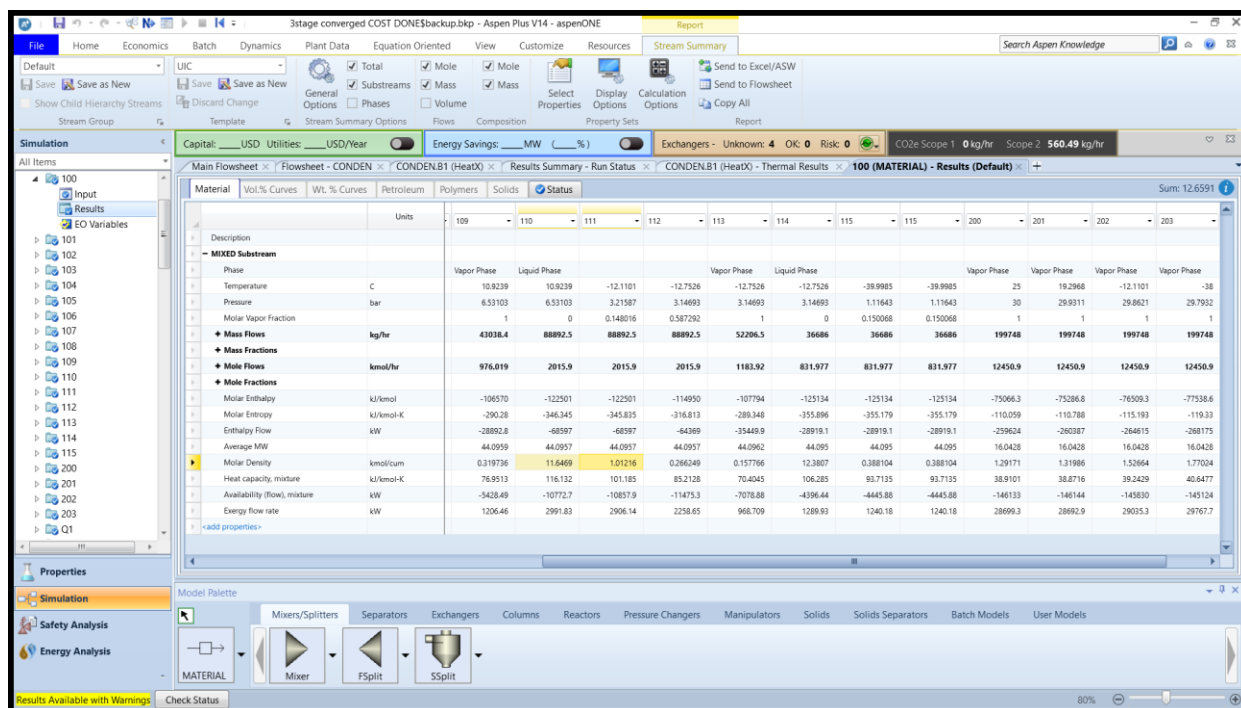
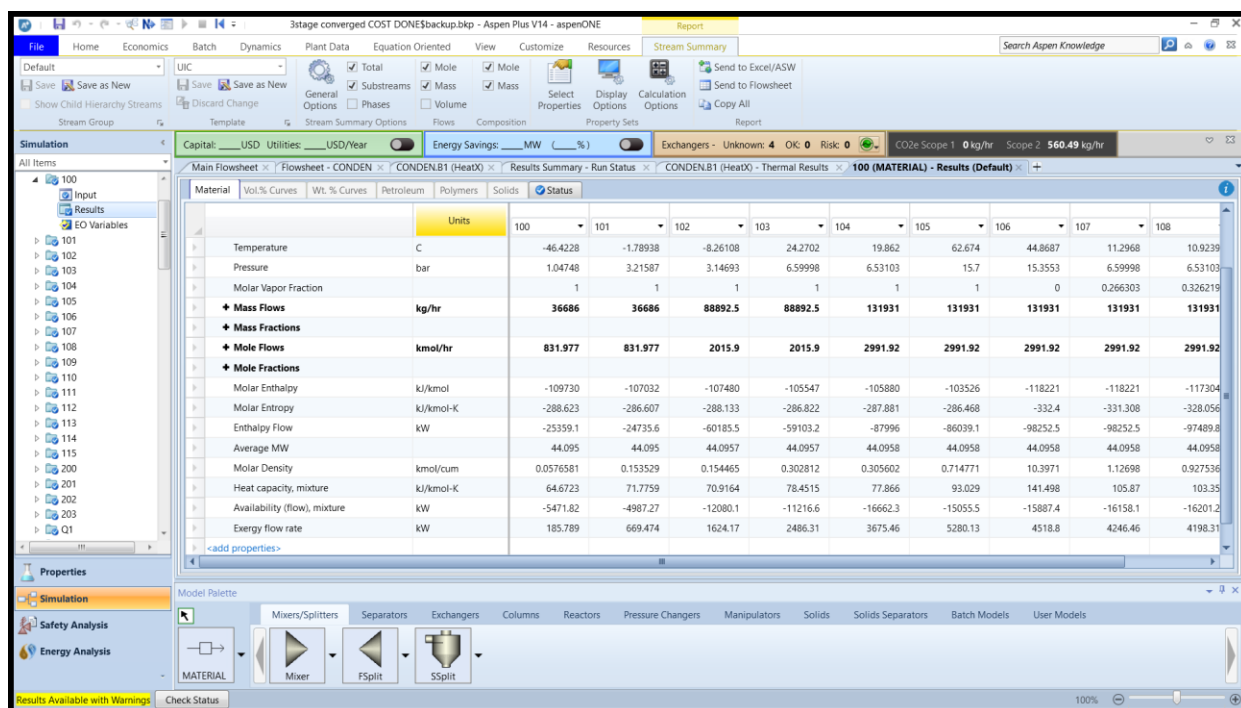


Figure 6 Heat and Weight Balance

2.3-Solution

To calculate the KPIs and all the costs three calculators were utilized. One called C-SIM held all the calculated temperature values while C-RES held the results like valve outlet

transfers and C-CALC which was used to calculate all the costs. Once all the necessary variables like heat exchanger area, valve flow rates, power and water cost were imported the given formulas in the problem statement were utilized to calculate the cost. All of the costs were multiplied by their corresponding lang factors and the given cost escalation factor. The Fortran code for the costs is as follows:

```

Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers - Uninc
Main Flowsheet - CONDEN - CONDEN.B1 (HeatX) Results Summary - Run Status - CONDEN.B1 (HeatX)
Define Calculate Sequence Tears Stream Flash Comments
Calculation method
Fortran Excel Neural Network Fortran Declarations
Enter executable Fortran statements
LFLV= 1.3
LFE= 2.5
LFX= 3.5
LFD= 4.0
CEF=1.5
ACOR=0.2
CWCOST= (ABS(QCOND)*.354*3600.0*8000)/1000000.0
OPEX=CWCOST+ELECOST
CPHK1= 28000 + 54.0*HK101A**1.20
CPHK2= 28000 + 54.0*HK102A**1.20
CPHK3= 28000 + 54.0*HK103A**1.20
CPCOND=28000 + 54.0*HKCOND**1.20
CK101= 580000 + 20000* W101**0.6
CK102= 580000 + 20000* W102**0.6
CK103= 580000 + 20000* W103**0.6
VLV101=8141*VLV101M**0.7
VLV102=8141*VLV102M**0.7
VLV103=8141*VLV103M**0.7
D103PUR=11600+ 34.0*2590.1**0.85
D102PUR=11600+ 34.0*5500.0**0.85
Row: 13 Col: 39

Calculation method
Fortran Excel Neural Network Fortran Declarations
Enter executable Fortran statements
VLV101=8141*VLV101M**0.7
VLV102=8141*VLV102M**0.7
VLV103=8141*VLV103M**0.7
D103PUR=11600+ 34.0*2590.1**0.85
D102PUR=11600+ 34.0*5500.0**0.85
CPHtot= CPHK1 + CPHK2+ CPHK3 +CPCOND
KINSTC=CPHKtot*LFX*CEF
CPEXtot=CK101 + ck102 + ck103
KINSTC=Cphtot*LFE*CEF
VLVtot=VLV101+ VLV102+ VLV103
VLVinstC= VLVtot*LFD*CEF
Dtot=D102pur+ d103pur
DinstC=Dtot*LFD*CEF
ACOR=KINSTC*ACOR
ACORV=KINST*ACOR
ACORU=DinstC*ACOR
ACORL=KINSTC*ACOR
CPEX=(ACORK + ACORV* ACOR + ACORL)/1000000
COT=CPEX*OPEX
Row: 13 Col: 39

```

Figure 7 Fortran Code

This code exports all the costs into the required table in terms of total cost and install costs of all the unit operations. For the mass of the drums the economic analysis tool was used to size the drum and that mass was imported into the given equations. To ensure the flowsheet could be optimized hierarchies were utilized for all the evaporators and condensers. Within every hierarchy a heat exchanger was placed that specified the minimum temperature difference from the hot and cold side using the given values in the problem statement as a guide starting from 10 all the way down to 2. For the final heat exchanger before the methane leaves the system the

outlet vapor fraction is specified to 0 since it is known that the methane is a saturated liquid. For all hierarchies the input streams are transferred using a transfer block from the main flowsheets ensuring the results are consistent. Doing this allows for further optimization without the heat exchangers causing errors in convergence. The KPIs are total power and cost is as follows:

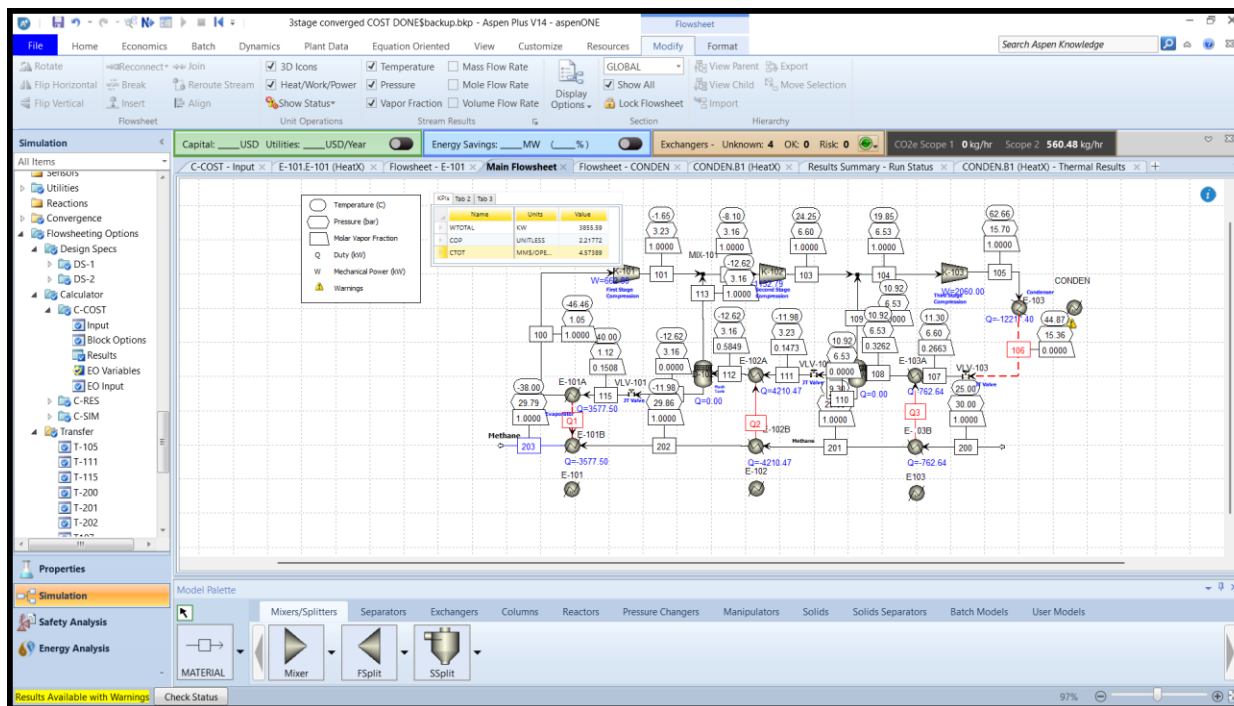
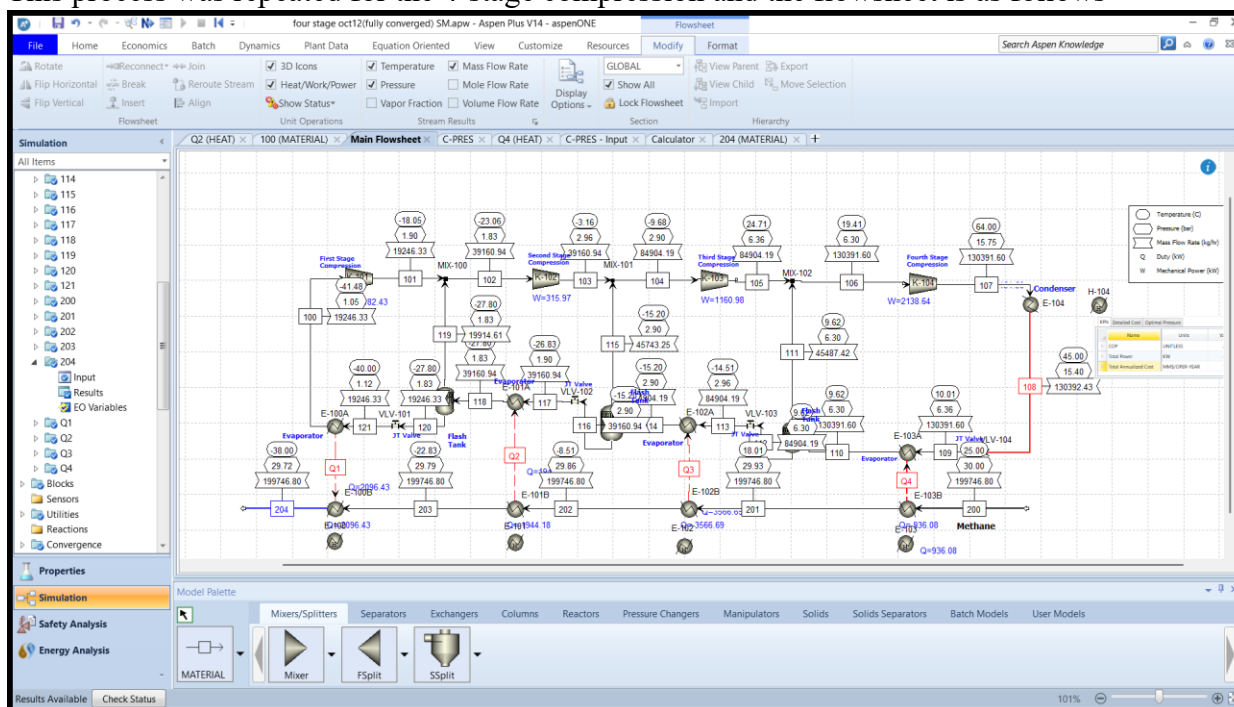


Figure 8 Final three stage

This process was repeated for the 4-stage compression and the flowsheet is as follows



The heat and weight balance for the 4 stage compression is as follows:

Main Flowsheet x Results Summary - Streams (All) x Results Summary x +																
Material	Heat	Load	Work	Power	Vol.% Curves	Wt.% Curves	Petroleum	Polymers	Solids							
	Units	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114
Description																
MIXED Substream																
Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Liquid Phase			Vapor Phase	Liquid Phase		
Temperature	C	-41.4765	-18.0504	-23.0636	-3.15813	-6.67753	24.7102	19.4063	63.9957	44.9995	10.0067	9.62334	9.62334	9.62334	-14.5111	-15.1959
Pressure	bar	1.04748	1.9	1.83105	2.96437	2.89542	6.36398	6.29503	15.7455	15.4008	6.36398	6.29503	6.29503	6.29503	2.96437	2.89542
Molar Vapor Fraction		1	1	1	1	1	1	1	1	1	0	0.275359	0.348854	1	0	0.152963
Mass Flows	kg/hr	19246.3	19246.3	39160.9	39160.9	84904.2	84904.2	130392	130392	130392	130392	130392	45487.4	84904.2	84904.2	84904.2
Mass Fractions																
Mole Flows	kmol/hr	436.459	436.459	888.073	888.073	1925.42	1925.42	2956.97	2956.97	2956.99	2956.97	2956.97	1031.55	1925.42	1925.42	1925.42
Mole Fractions																
Molar Enthalpy	kJ/kmol	-109401	-107971	-108303	-107086	-107538	-105476	-105880	-103406	-118200	-118200	-117060	-106633	-122647	-122647	-115978
Molar Entropy	kJ/kmol-K	-287.242	-286.112	-287.131	-286.224	-287.737	-286.34	-287.626	-286.147	-332.351	-331.174	-327.121	-290.243	-346.878	-346.323	-320.441
Enthalpy Flow	KW	-13263.6	-13090.3	-26716.9	-26416.7	-57515.5	-56412.6	-86967.4	-84935.7	-97087.8	-97087.2	-96151.1	-30554.8	-65596.3	-65596.3	-62029.6
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.0964	44.0964	44.0964	44.0964	44.0964	44.0962	44.0965	44.0965	44.0965
Molar Density	kmol/cum	0.0563098	0.0941937	0.0906712	0.141483	0.142071	0.289852	0.293568	0.711934	10.3913	1.0575	0.84231	0.308309	11.6891	0.913861	0.267902
Heat capacity, mixture	kJ/kmol-K	65.1222	68.5028	67.855	71.3629	70.4904	78.2617	77.553	93.1025	141.65	104.927	101.892	76.5224	115.483	100.238	86.2838
Availability (flow), mixture	KW	-2880.61	-2748.14	-5398.45	-5365	-11632.4	-10732.2	-16529.3	-14859.8	-15696.4	-15984.4	-16040.9	-5736.55	-10282.4	-10371	-10931.5
Energy flow rate	KW	85.4458	217.655	436.587	668.625	1452.73	2331.49	3566.26	5233.51	4467.27	4177.34	4114.6	1253.37	2861.22	2772.1	2165.97
<add properties>																
m Polymers Solids																

Figure 9 4 stage compression heat and weight balance

After comparison the optimal number of stages appears to be 4 stages as it has the lowest Power requirement and has the lowest Cto